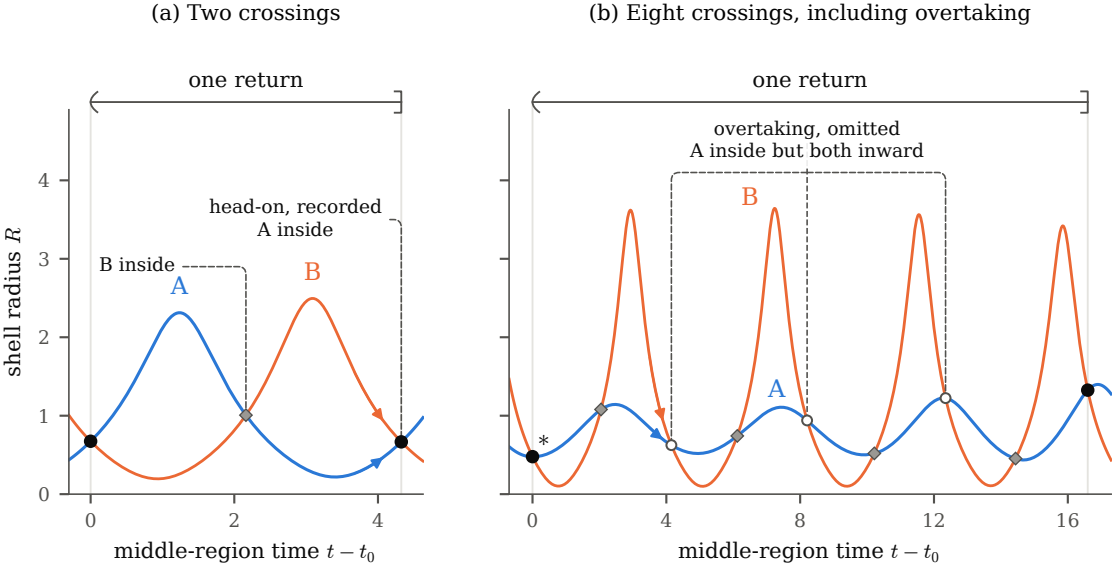
(a) Two crossings
(b) Eight crossings, including overtaking
one return
one return
shell radius R
head-on, recorded
A inside
B inside
A
B
overtaking, omitted
A inside but both inward
B
A
*
0
1
2
3
4
0
2
4
0
4
8
12
16
middle-region time $t - t_0$
middle-region time $t - t_0$